

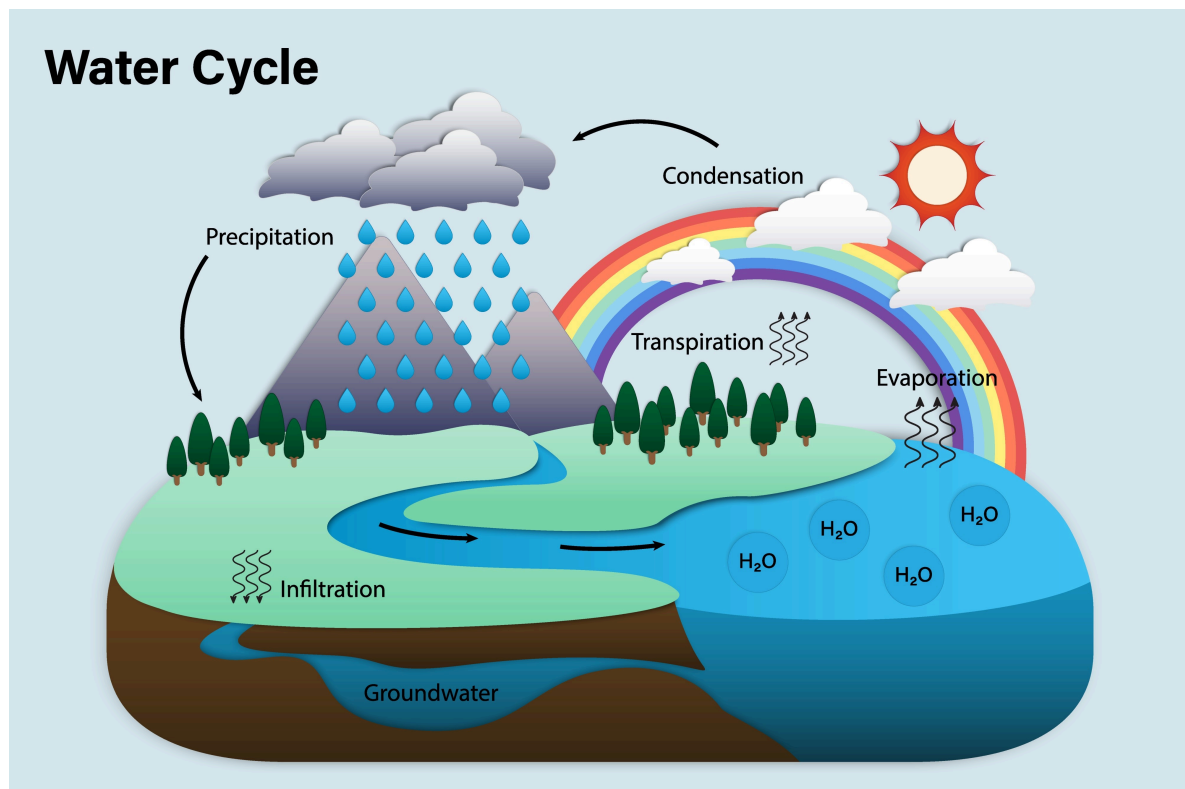
Week 7 MG.md

THE FINAL EXAM TRAP LIST (Review Before Exam)

- **The Quantity vs. Quality Trap:** Do not confuse these. **Quantity** = *how much* water you have. **Quality** = *how clean/safe* the water is.
- **The Flow Process Trap:** Disinfection happens **AFTER** filtration, never before. Intake structures are used for **surface water** (rivers/lakes), NOT groundwater.
- **The Nitrogen Age Trap:** The form of nitrogen tells you the age of the pollution. **Ammonia** = recent pollution. **Nitrite** = intermediate (should always be ZERO in safe water). **Nitrate** (NO_3^-) = fully oxidized, stable organic matter.

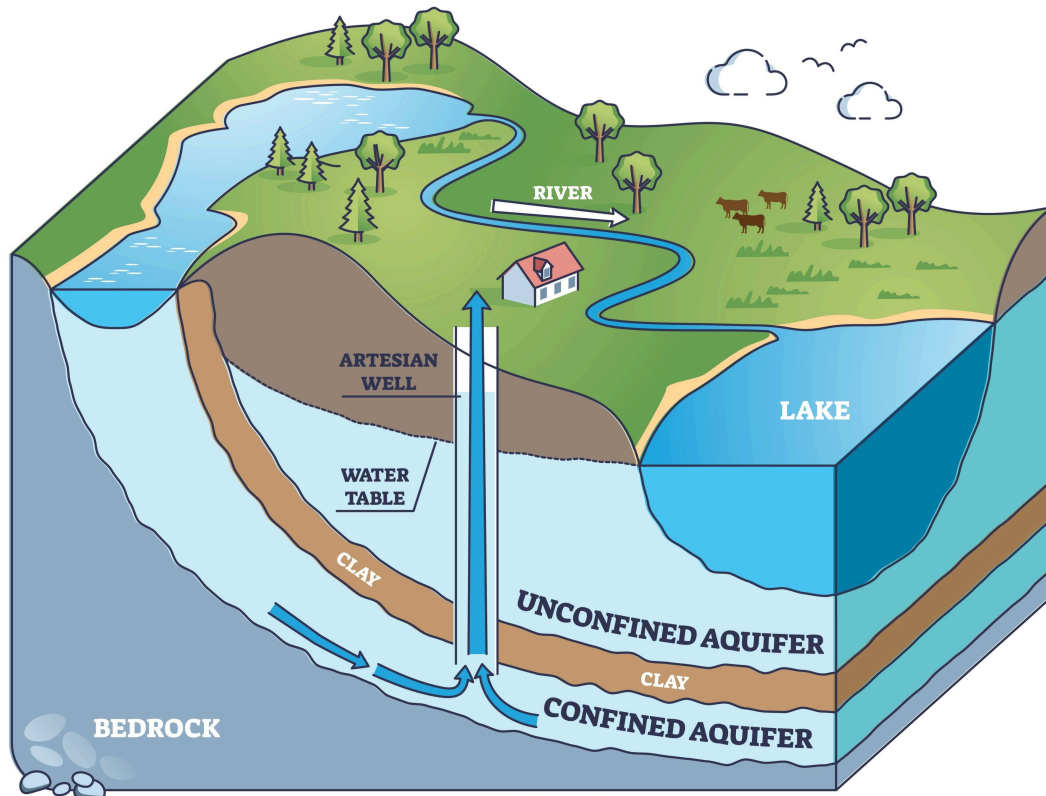
EVERY Week 7 Concept (Explained Like You're 5)

Part 1: Where Water Comes From (Hydrology)



1. **The Hydrologic Cycle:** The Earth's giant water wheel. Sun heats the ocean (**Evaporation**), plants sweat (**Transpiration**), clouds form (**Condensation**), and rain falls (**Precipitation**).
 - *Trap:* Units of water transfer globally are massive: $10^{12} \text{ m}^3/\text{year}$.
2. **Aquifers (The Underground Sponges):**

AQUIFER



- **Unconfined:** An open-top sponge. The top of the water is the “water table” and it sits at normal atmospheric pressure.
- **Confined:** A sponge squeezed between two solid rocks. The water is **under pressure**. If you poke a hole (Artesian Well), the water shoots up automatically without a pump!
- **3. The Water Supply Flow:** You must know the exact order: Reservoir → Intake (surface water only) → Treatment → Coagulation/Flocculation → Sedimentation → Filtration → **Disinfection** → Storage/Distribution. (Note: Disinfection is always AFTER filtration).

Part 2: Predicting the Future (Population & Demand)

- **4. The 4 Ways Cities Grow (Forecasting):**
 - **Arithmetic Method:** For old, packed cities (Grandpa city). Growth is a steady, constant number every 10 years ($P_n = P + n \cdot C$).
 - **Geometric Method:** For brand-new industrial towns (Baby city). Growth multiplies by a constant *percentage*.
 - **Incremental Method:** For average towns. It accounts for a growth rate that is slowly increasing.
 - **Logistic Curve:** For cities hitting their absolute maximum capacity (S-shaped curve).
- **5. The “Coincident Draft” (Pipe Sizing):** How big should the city’s water pipes be? You have to plan for the worst-case scenario. You design pipes for the MAXIMUM of either:
 - (Max Daily Demand + Fire Demand) **OR** (Maximum Hourly Demand).
 - *Rule of Thumb:* Max Daily = $1.8 \times \text{Average}$. Max Hourly = $2.7 \times \text{Average}$.

Part 3: Rain and Pavement (Runoff)

- **6. The Runoff Coefficient (C):** If rain falls on grass ($C = 0.15$), the dirt drinks it up. If rain falls on concrete pavement ($C = 0.80$) or a roof ($C = 0.90$), the water slides right off. The higher the number, the less water gets absorbed.
- **7. The Rational Method:** $Q = \frac{C \cdot I \cdot A}{360}$. This formula tells you exactly how much storm water will flood the streets, combining the pavement type (C), rain intensity (I), and city area (A).

Part 4: Water Quality – The Good, Bad, and Ugly

- **8. The Nitrogen Story (Time Machine):**
 - **Ammonia (NH_3):** Fresh poop. Very recent pollution.
 - **Nitrite (NO_2^-):** In the middle of rotting. **Highly dangerous** (Limit = Zero!).
 - **Nitrate (NO_3^-):** Old pollution. Safe for adults, but if it exceeds **45 mg/L**, it causes **Methemoglobinemia (Blue Baby Syndrome)** in infants by stealing oxygen from their blood.
 - **9. Fluoride (The Tooth Fairy):** * Perfect amount = ~ 1.0 ppm (stops cavities).
 - Too much (> 1.5 ppm) = **Fluorosis** (yellow spotting and weak teeth/bones).
 - **10. Hardness:** Calcium and Magnesium in the water. It prevents soap from lathering and clogs boilers with scale.
 - *Temporary:* Caused by carbonates/bicarbonates. You can fix it just by **boiling** the water.
 - *Permanent:* Caused by sulfates/chlorides. Boiling does nothing.
 - **11. Coliforms (The Poop Detectors):** We can't test for every disease, so we test for *E. coli* (coliforms) found in human intestines. Safe drinking water must have ≤ 1 **coliform per 100 mL**.
 - **Bacteria & Pathogen Rules:**
 - **Facultative Bacteria:** A specific group of bacteria that can survive with OR without free oxygen.
 - **Coliform Index:** Defined as the reciprocal of the smallest sample giving a positive result.
 - **Safe Coliform Level:** The standard is ≤ 1 coliform per 100 ml of water.
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The 100% COMPLETE Week 7 Q&A Bank

(Compiled from 2019, 2023, 2024, 2025, v2, formula sheet, and new PDFs)

Hydrology, Aquifers, and Runoff

- **Q:** Which of the following is NOT a component of the hydrologic cycle? **A:** Sedimentation (Components are Evaporation, Transpiration, Condensation, Precipitation, Runoff, Infiltration).
- **Q:** What is the pressure of the upper surface of an unconfined aquifer? **A:** Equal to atmospheric pressure.
- **Q:** Aquifers are distinguished into how many types on the basis of physical conditions? **A:** 2 (Unconfined and Confined).
- **Q:** Which type of land cover has the highest runoff coefficient? **A:** Roofs (0.90) and Pavements/Yards (0.80).
- **Q:** Which type of land cover will have less runoff coefficient? **A:** Open ground (0.10) or Lawns/Gardens (0.15).
- **Q:** Calculate the quantity of storm water runoff: Area = 300 hectares, Intensity = 30 cm/h, overall runoff coefficient = 0.44.

- **A:** 110 m³/sec.
- | *Note:* $Q = CiA = 0.44 \times (30 \text{ cm/h}/3600 \text{ s}) \times (300 \times 10^4 \text{ m}^2) = 110 \text{ m}^3/\text{s}.$
- **Q:** Calculate stormwater runoff: $I=4\text{mm/min}$, $C=0.6$, $A=400\text{ha}$.
 - **A:** 160 m³/sec.
 - | *Note:* $I = 4 \text{ mm/min} = 240 \text{ mm/h}.$
 $Q = CIA/360 = (0.6 \times 240 \times 400)/360 = 160.$
- **Q:** Which land cover type has a runoff coefficient of approximately 0.50? **A:** Single-family dwellings.
- **Q:** In the rational method ($Q = CIA/360$), the intensity I used must be for a duration equal to: **A:** The time of concentration of the catchment.
- **Q:** If $C = 0.80$, $I = 45 \text{ mm/hr}$, $A = 100 \text{ ha}$, what is the runoff Q ? **A:** 10 m³/sec.
- **Q:** If $C = 0.63$, $I = 40 \text{ mm/hr}$, $A = 200 \text{ ha}$, what is the runoff Q ? **A:** 14 m³/sec.

Water Demand & Population Forecasting

- **Q:** Which demand governs the design of distribution pipes in dense urban areas? **A:** Fire demand + maximum daily demand (The Coincident Draft).
- **Q:** Maximum daily demand is typically how many times the average daily demand? **A:** 1.8 times.
- **Q:** Identify the formula for calculating the maximum hourly water demand. **A:** $2.7 \times \text{Average daily demand}$ (or $1.5 \times \text{Max daily demand}$).
- **Q:** Fire demand is characterized by: **A:** Small annual volume and high discharge rate.
- **Q:** Calculate the fire demand of a city for a population of 10,00,000 using National Board of Fire Underwriters formula? **A:** ~145 million liters/day (Based on $Q = 4637\sqrt{P}(1 - 0.01\sqrt{P})$ where P is in thousands).
- **Q:** Which population forecasting method is suitable for a large and old city with stable growth? **A:** Arithmetic mean / Arithmetical increase method.
- **Q:** Which population forecasting method is suited for a new industrial town at the beginning of development? **A:** Geometrical increase method.
- **Q:** Which population method assumes a constant rate of increase? **A:** Arithmetical Increase Method ($P_n = P + n \cdot C$).
- **Q:** If the population increased by 18k, 22k, and 20k over 3 decades. Present is 300k. Population after 2 decades (Arithmetic)?
 - **A:** 3.40 lakh.
 - | *Note:* $\text{Average increase} = 20\text{k}.$ $300\text{k} + 2(20\text{k}) = 340\text{k}.$
- **Q:** What factor does NOT affect per capita water demand? **A:** Type of soil in the area. (Factors that DO affect it: Cost of water, climate, city size, metering).
- **Q:** Which population forecasting method divides the city into zones with fixed population densities as per a 25–30 year city plan? **A:** Master Plan Method.
- **Q:** Fire demand formulas (like Kuichling's and Freeman's) express fire demand in what units? **A:** Litres/min.
- **Q:** If the population in 2021 is 2,000,000 and grows at 1.5% annually, what is the population in 2031 using the Geometrical Increase Method? **A:** 2,030,000.
- **Q:** If the average increment is 8,000 and the incremental increase is 1,000, what is the population in 2021 using the Incremental Increase Method, given the 2011 population was 28,000? **A:** 37,000.

Pipe Flow (Darcy & Hazen-Williams)

- **Q:** A town of 1 lakh, 80 km from source, rate 200 lpcd. Find pipe diameter if $f = 0.003$ and elevation difference = 10m.
 - **A:** 0.64 m.
 - **Note:** $Q = 0.231 \text{ m}^3/\text{s}$. Darcy-Weisbach: $h_f = \frac{fLQ^2}{12.1D^5}$.
 $10 = \frac{0.003 \times 80000 \times 0.231^2}{12.1D^5} \rightarrow D = 0.64 \text{ m}.$
- **Q:** Using Hazen-Williams, estimate head loss: $D = 0.3 \text{ m}$, $L = 5 \text{ m}$, $Q = 1.2 \text{ m}^3/\text{s}$, $C_h = 130$.
 - **A:** 3.2 m.
 - **Note:** $h_f = \frac{10.67 \times L \times Q^{1.852}}{C_h^{1.852} \times D^{4.87}}$.
- **Q:** A pipeline delivers water 5 km away. Pipe diameter = 0.4 m, $f = 0.002$, $v = 3 \text{ m/s}$. Determine head loss using Darcy-Weisbach. **A:** 11.5 m.
- **Q:** A water supply plant is designed for a city having a population of 1,50,000. Estimate the total draft for an average consumption of 250 Lpcd. Given fire demand is 71.8 MLD.
 - **A:** 139.3 MLD. (Math: To find Max Daily Demand, multiply average by 1.8.
 $1.8 \times 150,000 \times 250/10^6 = 67.5 \text{ MLD}$. Total Draft = $67.5 + 71.8 = 139.3 \text{ MLD}$).

Water Quality & Diseases

- **Q:** Which form of nitrogen indicates fully oxidized organic matter (old pollution)? **A:** Nitrate.
- **Q:** Which nitrogen indicates the very first stage of decomposition (recent pollution)? **A:** Free Ammonia.
- **Q:** Methemoglobinemia (Blue Baby Syndrome) is mainly caused by: **A:** Nitrates in water (> 45 mg/L).
- **Q:** What is the limiting concentration of fluoride in drinking water? **A:** 1.5 mg/L. (Excess causes Fluorosis).
- **Q:** Name the group of bacteria that can survive with or without free oxygen. **A:** Facultative bacteria.
- **Q:** Which instrument is used to measure the odour of water? **A:** Osmoscope.
- **Q:** Calculate the Threshold odour number (TON) of the sample if 50 ml water sample is diluted with 250 ml distilled water at which odour is just detectable.
 - **A:** 6. (Math: $\text{TON} = (A + B)/A \rightarrow (50 + 250)/50 = 6$).
- **Q:** A 100 mL sample has an initial wt of 34.563g, and Final wt of 34.589g. Calculate total solid concentration.
 - **A:** 260 mg/L. (Math: $\text{Mass} = 34.589 - 34.563 = 0.026 \text{ g} = 26 \text{ mg}$. $\text{Concentration} = 26 \text{ mg}/0.1 \text{ L} = 260 \text{ mg/L}$).
- **Q:** Calculate the Threshold Odour Number (TON) if 50 ml water is diluted with 250 ml distilled water at which odour is just detectable.
 - **A:** 6.
 - **Note:** $\text{TON} = (A + B)/A = (50 + 250)/50 = 6$.
- **Q:** If the pOH of the water sample is 11, what is the nature of the water? **A:** Acidic ($\text{pH} = 14 - 11 = 3$).
- **Q:** "If the oxygen present in water is found to be greater than its saturation level, it indicates presence of organic matter." **A:** False (Supersaturation indicates algae/photosynthesis. Organic matter lowers DO).
- **Q:** Which salts indicate permanent hardness of water? **A:** Calcium sulphate (Sulfates and Chlorides of Ca/Mg).

- **Q:** Identify the physical properties of water. **A:** Turbidity, Colour, Total Solids.
- **Q:** Below what fluoride concentration does dental cavity (tooth decay) occur? **A:** 0.8–1.0 ppm.
- **Q:** Which metal, if exceeded (>0.3 ppm), causes discolouration of clothes? **A:** Iron (Fe).
- **Q:** High concentrations of sulphate in drinking water can cause what health effect when combined with calcium and magnesium? **A:** Laxative effect.
- **Q:** If a water sample has a pH of 8, what is its hydroxide ion concentration $[\text{OH}^-]$? **A:** 10^{-6} mol/L.
- **Q:** Which pH indicator changes from yellow to red in the range 6.8–8.4? **A:** Phenol red.

Hardness & Alkalinity Calculations

- **Q:** Cations: $\text{Na}^+=20$, $\text{Ca}^{2+}=60$, $\text{Mg}^{2+}=36$. Compute total hardness as CaCO_3 .
 - **A:** 300 mg/L.
 - **Note:**

$$\text{TH} = (\text{Ca} \times 50/20) + (\text{Mg} \times 50/12.2) = (60 \times 2.5) + (36 \times 4.1) = 150 + 150 = 300.$$
- **Q:** $\text{HCO}_3^- = 244$ mg/L. Compute total alkalinity as CaCO_3 .
 - **A:** 200 mg/L.
 - **Note:** $\text{Alk} = \text{HCO}_3^- \times (50/61) = 244 \times 0.82 = 200.$
- **Q:** Compute carbonate hardness (CH) and non-carbonate hardness (NCH) if TH = 300 and Alk = 200.
 - **A:** CH = 200 mg/L, NCH = 100 mg/L.
 - **Note:** CH = Minimum of (TH, Alkalinity). NCH = TH - CH.
- **Q:** Alkalinity in water is caused by the presence of: **A:** Bicarbonates and carbonates of calcium/magnesium, or hydroxides of sodium/potassium.
- **Q:** A water sample has 100 mg/L of Ca^{2+} and 120 mg/L of Mg^{2+} . Calculate the total hardness as CaCO_3 . **A:** 750 mg/L. (Note: The Master Guide only had the 300 mg/L variant).
- **Q:** The catchment area is 500 hectares. Calculate the quantity of storm water runoff, if intensity of rainfall is 10 cm/h, Overall runoff coefficient is 0.62?
 - **A:** 86 m^3/s . (Math:

$$Q = C \times I \times A \rightarrow 0.62 \times (0.1 \text{ m}/3600 \text{ s}) \times (500 \times 10,000 \text{ m}^2) = 86.11 \text{ m}^3/\text{s}.$$

ACTIVE RECALL & FEYNMAN CHALLENGE

Cover the answers below and test yourself before the exam.

Quiz 1: If a factory wants to calculate the exact max daily demand for a new water treatment plant, why must they multiply the average daily demand by 1.8?

Answer: Because water usage is not perfectly even throughout the year. The “1.8” multiplier accounts for the maximum daily demand during peak seasons (like summer) to ensure the plant doesn’t run dry on the hottest day of the year.

Quiz 2: You have a water sample where the Total Hardness (TH) is 250 mg/L as CaCO_3 , and the Total Alkalinity is 350 mg/L as CaCO_3 . What is the Non-Carbonate Hardness (NCH)?

Answer: 0 mg/L. The rule is that Carbonate Hardness (CH) equals the minimum of TH or Alkalinity. Since TH (250) is smaller, $\text{CH} = 250$. Since $\text{NCH} = \text{TH} - \text{CH}$, $250 - 250 = 0$.

Feynman Challenge: Explain the difference between an Unconfined Aquifer and a Confined Aquifer using a household sponge and a plastic bag.

Example Answer: Imagine a wet household sponge sitting on your kitchen counter. If you pour water on it, it soaks right in, and the top of the water inside the sponge feels the normal air pressure of the room. That is an Unconfined Aquifer. Now, take that same wet sponge, put it inside a thick plastic zip-lock bag, put a heavy book on top of it, and zip it shut. The water is trapped and under immense pressure. If you poke a hole in the bag with a straw (an artesian well), the water will shoot up through the straw on its own because of the pressure. That is a Confined Aquifer!